

Chapter 9: The number plane

Drill practice and mixed revision of all 6 lessons · 53
questions

NSW syllabus outcome

Stage 4 Number and algebra: MA4-LIN-C-01 creates and
displays number patterns and graphs linear relationships on the
Cartesian plane.

CHAPTER

9



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 9. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–28

9.01 Grid references on maps

Easy Write the grid reference: the column letter first, then the row number.

① column F, row 5

② column A, row 3

③ column D, row 3

9.02 Points in the first quadrant

Easy Write the x-coordinate (x) or the y-coordinate (y) of the point.

④ y of (2, 2)

⑤ x of (5, 10)

⑥ x of (12, 6)

⑦ x of (8, 8)

Medium The two points are on the same horizontal or vertical line. How far apart are they?

⑧ (3, 2) and (3, 9)

⑨ (5, 4) and (5, 3)

⑩ (2, 3) and (2, 10)

9.03 The number plane

Easy Which quadrant (1st, 2nd, 3rd or 4th) or which axis is the point on?

⑪ (8, 2)

⑫ (2, -8)

⑬ (3, 9)

⑭ (8, -2)

Medium Reflect the point in the axis given.

⑮ (-6, 7) in the x-axis

⑯ (-8, 6) in the y-axis

⑰ (-4, -7) in the x-axis

⑱ (-7, -4) in the x-axis

9.04 Plotting tables of values

Easy Write each column of the table as a point (x, y).

⑲ x: 0, 1, 2, 3

y: 2, 3, 4, 5

⑳ x: 0, 1, 2, 3

y: 3, 4, 5, 6

Medium Describe the pattern: what happens to y each time x goes up by 1?

㉑ (0, 7), (1, 12), (2, 17)

㉒ (0, 6), (1, 5), (2, 4)

㉓ (0, 7), (1, 4), (2, 1)

9.05 Tables of values from rules

Easy Substitute to find y.

㉔ $y = 3x - 3$, $x = 5$

㉕ $y = 6x + 2$, $x = 1$

㉖ $y = 3x + 4$, $x = 2$

Medium Find the rule, $y = \dots x + \dots$

㉗ x: 0, 1, 2, 3

y: 6, 7, 8, 9

㉘ x: 0, 1, 2, 3

y: 2, 6, 10, 14

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 29–35

9.06 Graphing linear equations

Easy Does the point lie on the line? Write yes or no.

29 $(4, 15)$ on $y = 3x + 3$

30 $(-1, 2)$ on $y = x + 1$

31 $(-2, -8)$ on $y = 3x - 3$

Medium The point lies on the line. Find the missing coordinate.

32 $(-1, ?)$ on $y = 3x + 3$

33 $(-3, ?)$ on $y = -2x + 5$

34 $(-2, ?)$ on $y = -3x - 1$

35 $(1, ?)$ on $y = -3x + 4$

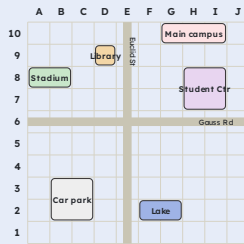
YOU DO On your own: Part 2, mixed revision.

QUESTIONS 36–45

36 Easy

Use the campus map. What is at the grid reference?

I10 Lesson 9.01



38 Easy

Which quadrant is the point in?

$(-6, -2)$ Lesson 9.03

40 Easy

Complete each table.

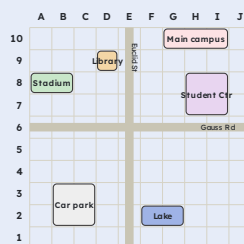
$d = 5 - c$ Lesson 9.05

c	-1	0	1	2	3
d					

42 Medium

Use the map.

Which features are in column H? Lesson 9.01



37 Easy

Write the x-coordinate.

x of $(5, 2)$ Lesson 9.02

39 Easy

Write the column as a point (x, y) .

$x = -3, y = 2$ Lesson 9.04

41 Easy

Does the point lie on $y = 2x + 1$?

$(4, 9)$ Lesson 9.06

43 Medium Where is the point $(0, 5)$? Lesson 9.02

44 Medium Write the coordinates of a point in the 4th quadrant. Lesson 9.03

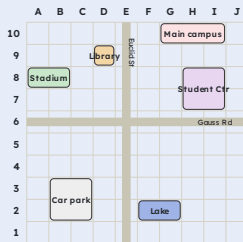
45 Medium Describe the pattern of the points $(1, 4), (2, 4), (3, 4)$. Lesson 9.04

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 46–53

- 46** **Medium** For $y = 4x - 1$, complete for $x = 0, 1, 2$.
Lesson 9.05

- 48** **Challenging**
Draw your own.
Write directions from one place to another using grid references. Lesson 9.01



- 50** **Challenging** Reflect $(-2, 5)$ in the x -axis. Lesson 9.03

- 52** **Challenging** Find the rule: $x = 0, 1, 2$ and $y = 1, 4, 7$.
Lesson 9.05

- 47** **Medium** Does $(2, 5)$ lie on $y = 2x + 1$? Lesson 9.06

- 49** **Challenging** A rectangle has corners $(1, 2)$, $(6, 2)$, $(6, 5)$. Find the fourth corner. Lesson 9.02

- 51** **Challenging** What happens to y each time x goes up by 1 for $(0, 1)$, $(1, 4)$, $(2, 7)$? Lesson 9.04

- 53** **Challenging** Does $(-1, 3)$ lie on $y = -2x + 1$?
Lesson 9.06

YOU DO

On your own: extra working space. Write the question number next to your working.